

[CONFIDENTIAL LAB NOTEBOOK - OPTICS LAB EXP #4]

Date: October 14, 2026. Station: Michelson Interferometer & Diffraction Grating.

Observer: Dr. K. Vance. Room temp: 21.4 C. Pressure: 101.3 kPa.

1. Diffraction Grating Equation and Orders of Interference

Light from a 632.8 nm He-Ne laser incident on a grating with $d = 1.667$ micrometers (600 lines/mm).

Principal maxima condition:

$$d \cdot \sin(\theta_m) = m \cdot \lambda$$

where m is the integer diffraction order ($m = 0, \pm 1, \pm 2$).

2. Experimental Measurement Data Log

Order m | Measured Angle (deg) | Theoretical Angle (deg) | Discrepancy

+1	22.31	22.31	0.00%
+2	49.44	49.42	+0.04%
-1	22.33	22.31	+0.09%
-2	49.39	49.42	-0.06%

Figure 1 (Page 1): Handwritten sketch of spectrometer detector sweep and laser photodiode alignment.

Notes: Grating optical axis re-calibrated. Zero-order reflection beam blocked by stop.